Chakrayuddh Kimsovanna

Graduate Student | Data Analytics & Economics | Stata, R, SQL, Excel
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Graduate student in social data analytics and economics with experience in statistical modeling, machine learning, and applied economic research. Skilled in R, Stata, and SQL for analyzing large datasets and conducting empirical research.

EDUCATION

The University of Texas at Dallas — Expected July 2026

- *Masters of Science, Social Data Analytics and Research* — **GPA 3.945**

The University of Texas at Dallas — December 2024

- *Bachelor of Science, Economics,* Minor: International Political Economy — **GPA 3.781**

PROJECTS

Analyzing H-1B Visa Sponsorship Trends in Finance-Related Occupations (2016-2024) Using Text Mining — March-May 2025

Paper: [Link] | Github: [Link]

- Processed 2.5 GB of USCIS and DOL data in R/Python, applying BERT embeddings (MiniLM-L6-v2) and K-Means clustering to classify finance-related occupations.
- Grouped job titles into 8 semantic clusters, revealing wage and demand trends from 2016–2024.
- Delivered insights through data visualizations and text analytics, highlighting shifts across policy periods.

Designing a Database Application for Cambodia's FDI & Trade — March-May 2025

Interactive App: [Link] | Github: [Link]

- Built a PostgreSQL/SQLite relational database integrating investment and trade data (Open Development Cambodia, World Bank WITS) for financial pattern analysis.
- Developed an interactive R Shiny dashboard visualizing FDI inflows by province, sector, and investor country using Leaflet maps and pie charts.
- Published a functional web app and SQL pipeline improving data validation and access for international trade and investment reporting.

Evaluating Minimum Wage and Labor Force Participation in the U.S. (2006–2025) — October-December 2025

Presentation: [Link] | Interactive App: [Link] | Github: [Link]

- Collaborated in a 3-person team to construct a state-level panel dataset using FRED and Census APIs, integrating minimum wage, LFPR, and socio-economic controls in R
- Applied OLS and fixed-effects regression models to evaluate the relationship between minimum wage and labor force participation
- Developed an interactive Shiny dashboard to visualize state-level trends and highlight structural drivers of labor market participation.

ReCell: Predictive Modeling for Resale Prices of Consumer Electronics — January – February 2024

Project: [Link]

- Built a regression model in Python (pandas, scikit-learn) explaining approximately 84% of price variation, achieving predictions within ~4.5% error on test data
- Identified key pricing drivers (e.g., device age, camera resolution, new device price) by estimating marginal effects and interpreting model coefficients
- Generated data-driven recommendations to support pricing and inventory decisions in the refurbished electronics market.

PRESENTATIONS

- "Designing a Database for Cambodia's Foreign Direct Investment and Trade," Graduate Research Project Presentation, M.S. Social Data Analytics and Research, University of Texas at Dallas, 2025.
- "Analyzing H-1B Visa Sponsorship Trends in Finance-Related Occupations (2016-2024) Using Text Mining," Graduate Research Project Presentation, University of Texas at Dallas, 2025.
- "The Relationship Between Effective Minimum Wage and State Labor Force Participation in the U.S. (2006-2025)," Graduate Research Project Presentation *(group project)*, University of Texas at Dallas, 2025.

ADDITIONAL INFORMATION

- **Analytical & Technical Skills:** Stata, R (tidyverse, ggplot2, dplyr), SQL (PostgreSQL, SQLite), Python, LaTeX, Excel
- **Data Analysis & Modeling:** Regression Analysis, Forecasting, Predictive Modeling, Statistical Inference, Data Visualization
- **Productivity Tools:** Microsoft Office Suite (Excel, PowerPoint, Word), Google Workspace (Docs, Sheets, Slides), GitHub

- **Relevant Coursework:** Econometrics, Business & Economic Forecasting, Mathematical Economics, Knowledge Mining, Information Management, Descriptive & Inferential Statistics, Regression & Multivariate Analysis
- **Work Authorization:** Eligible for employment under Curricular Practical Training (CPT); no visa sponsorship required

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CERTIFICATIONS

- Certificate in Data Science and Business Analytics – *The University of Texas at Austin, McCombs School of Business* (May 2024) — focused on data analysis, statistics, and machine learning with Python

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AWARDS & HONORS

- Cum Laude, B.S. in Economics, University of Texas at Dallas, 2024
- Dean's List, Fall 2024